2.REQUIREMENT ANALYSIS PHASE

Introduction:

The Requirement Analysis Phase focuses on identifying and documenting the requirements needed to develop the *EventForce Management System*. This phase helps in understanding the needs of event organizers and participants and defines the functions and resources required for the successful development of the system.

The requirements are divided into *functional requirements, which describe what the system should do, and **non-functional requirements*, which describe the quality and performance expected from the system.

Functional Requirements:

The functional requirements of the EventForce Management System include the following:

User Registration and Login

* Users should be able to create an account using their basic details.

* Registered users should be able to securely log in to the system.

* The system should validate user credentials before granting access.

Event Management

* Administrators should be able to create new events.

* Administrators should be able to update or delete event information.

* The system should store event name, description, date, time, venue, and other relevant details.

* Users should be able to view available events.

Event Registration

* Users should be able to register for available events.

* The system should store registration details.

* The system should prevent duplicate registrations for the same event.

* Users should be able to view their registered events.

Participant Management

* Administrators should be able to view registered participants.

* Participant details should be maintained systematically.

* Administrators should be able to monitor the number of registrations for each event.

Event Scheduling

* Administrators should be able to assign dates, times, and venues for events.

* Users should be able to view event schedules.

* The system should display updated scheduling information.

Notifications and Updates

* The system should provide important event-related updates to users.

* Participants should be informed about changes to event details whenever applicable.

Dashboard and Reports

* Administrators should have access to a dashboard containing event and registration information.

* The system should display registration statistics.

* Administrators should be able to generate or view organized event and participant records.

Non-Functional Requirements:

Usability

The system should have a simple, clear, and user-friendly interface so that users can easily navigate and perform required activities.

Performance

The system should respond quickly to user requests and efficiently handle event and registration data.

Security

User information and login credentials should be protected. Only authorized users should be allowed to access administrative functions.

Reliability

The system should provide consistent and accurate results during event creation, registration, and data management.

Scalability

The system should be designed so that additional events, users, and features can be added in the future without major changes to the existing system.

Maintainability

The system should have a well-structured design that makes it easier to maintain, update, and improve.

User Requirements:

Administrator

The administrator should be able to:

* Log in securely.
* Create and manage events.
* Update event details.
* Manage participants.
* Monitor registrations.
* View event statistics and reports.

Participant

The participant should be able to:

* Register and log in.

* Browse available events.

* View event details.

* Register for events.

* View registered events.

* Receive important event updates.

Expected Outcome of Requirement Analysis:

At the end of the Requirement Analysis Phase, all major functional and non-functional requirements of the EventForce Management System will be clearly identified and documented. These requirements will serve as the foundation for the *System Design and Development Phases*, ensuring that the final system meets the needs of both event organizers and participants.